L3b: Equity-Price Lattices Under Real-World and Risk-Neutral Measures

CHEME 5660 Cornell University

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Objectives for Today

Today we replace continuous price uncertainty with a finite lattice, derive its node distribution, and ask which probability law answers which question.

Today's concepts

- Build a binomial lattice: derive node prices and node probabilities from the up factor, down factor, horizon, and transition probability.
- Separate probability measures: explain why real-world probabilities support forecasting while risk-neutral probabilities support valuation.
- Test model purpose: evaluate probability of profit under the real-world measure and identify stylized facts excluded by fixed parameters.

Companion notes: L3b lecture notes and notebooks

Lectures and Examples

Lecture: CHEME-5660-L3b-Lecture-LatticeModels-Equity-Price-Fall-2026.ipynb

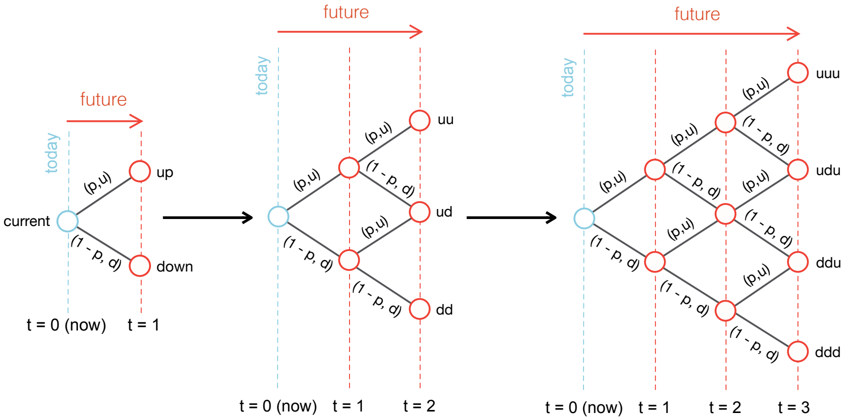
Example:
CHEME-5660-L3b-LatticeModelSharePrice-RWPM-Example-Fall-2026.ipynb

The example checks a textbook tree, estimates asymmetric move factors and a physical up probability from historical data, populates a recombining lattice, and compares its prediction bands with observed prices.

The factors are not forced to satisfy $d = 1/u$. That restriction is unnecessary for recombination.

Link: https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git

A Recombining Price Lattice



1-step ahead prediction

2-step ahead prediction

3-step ahead prediction

Binomial Lattice Model

Let $S_0 > 0$ be today's ex-dividend share price. Over each step of length $\Delta t > 0$ years, the price is multiplied by a constant up factor $u > 0$ or a constant down factor $d > 0$, with $d < u$.

Under a real-world up probability $p \in [0, 1]$, the one-step price is:

$$S_1 = \begin{cases} S_0 u, & \text{with probability } p, \\ S_0 d, & \text{with probability } 1 - p. \end{cases}$$

The fixed-parameter model assumes the same u, d, and p at every node and independent move draws across steps. Level n contains $n + 1$ price states.

Purpose: the lattice replaces a continuum of possible prices with a finite map whose probabilities and payoffs can be computed exactly.

Terminal Prices and Their Distribution

Let n be the number of independent steps, and let K_n count the up moves. Under the real-world law, $K_n \sim \mathrm{Binomial}(n, p)$. Conditional on $K_n = k$, where $k \in \{0, \dots, n\}$, the terminal price is:

$$\underbrace{S_{n,k} = S_0 u^k d^{n-k}}_{\text{terminal price}}.$$

There are $\binom{n}{k}$ paths with that count, and every such path has probability $p^k(1-p)^{n-k}$. The node probability is therefore:

$$\underbrace{\mathbb{P}(K_n = k) = \binom{n}{k} p^k (1-p)^{n-k}}_{\text{probability of node } (n,k)}.$$

The expected one-step gross return under this law is $pu + (1-p)d$.

Paths Grow Exponentially, Nodes Grow Linearly

After n binary moves, the unmerged path tree contains 2^n paths to the terminal level. The recombining lattice contains only $n+1$ terminal price nodes.

Why the compression is exact

- Every path with k up moves and $n-k$ down moves reaches $S_0 u^k d^{n-k}$.
- The binomial coefficient $\binom{n}{k}$ records how many path orders merge into that node.
- Recombination preserves the terminal distribution while replacing path enumeration with one state per up-move count.

This state compression is what makes backward lattice calculations tractable.

Real-World Parameter Estimation

Let $g_2, \dots, g_T$ be historical growth rates sampled at interval $\Delta t > 0$ years. Initialize an up-factor set U and a down-factor set D.

1. For every observation, place $e^{g_t \Delta t}$ in U when $g_t > 0$, in D when $g_t < 0$, and skip the factor when $g_t = 0$.
2. Estimate the conditional move magnitudes:

$$\hat{u} = \frac{1}{|U|} \sum_{v \in U} v, \qquad \hat{d} = \frac{1}{|D|} \sum_{v \in D} v.$$

3. Estimate the up probability by $\hat{p} = |U|/(T-1)$, so zero-change days remain in the denominator.

The sample must contain at least one up move and one down move.

Historical Estimates Are Model Choices

The procedure turns historical observations into a real-world forecast law. It does not make the parameters timeless or uniquely correct.

- Averaging positive and negative move factors compresses a continuous return distribution into two representative outcomes.
- Constant u, d, and p discard serial dependence, changing volatility, and tail structure.
- Estimates inherit sampling error, the chosen observation window, data adjustments, and regime dependence.
- Validation must compare the entire out-of-sample distribution and path behavior with observations, not only the mean path.

Interpretation: $\hat{p}$ is an estimated physical frequency, not a risk-neutral pricing probability.

Replacement, Recombination, and Reciprocal Moves

Three statements describe different features of the model.

- With replacement: every step draws again from the same move set $\{u, d\}$. An up move does not remove the possibility of another up move.
- Recombining: move order does not change the final price because multiplication commutes:

$$\underbrace{S_0 ud}_{\text{up then down}} = \underbrace{S_0 du}_{\text{down then up}} .$$

- Reciprocal moves: $d = 1/u$, or $ud = 1$, makes up and down increments symmetric in log space. It is an additional parameter choice.

Vocabulary: the unmerged path structure is a tree. After equal-price states merge, the lattice is a directed acyclic graph in the strict graph-theoretic sense.

Example: Asymmetric Moves Still Recombine

The companion notebook first checks $S_0 = 20$, $u = 1.1$, $d = 0.9$, and $p = 0.6523$. These factors are asymmetric because:

$$d = 0.9 \neq \frac{1}{1.1} \approx 0.9091, \qquad ud = 0.99 \neq 1.$$

The two mixed paths still reach one shared node:

$$\underbrace{20(1.1)(0.9)}_{\text{up then down}} = \underbrace{20(0.9)(1.1)}_{\text{down then up}} = 19.8.$$

Each ordered path has probability $p(1-p)$, so the recombined middle node has probability $2p(1-p)$. Reciprocal factors are unnecessary for both the shared price and the binomial node probability.

Notebook: CHEME-5660-L3b-LatticeModelSharePrice-RWPM-Example-Fall-2026.ipynb

Two Measures, Two Questions

The symbol attached to “probability of up” must identify its purpose.

Real-world $\mathbb{P}$

- Up probability p is estimated or modeled from data.
- Used for forecasts, scenarios, risk, and probability of profit.
- Tested against out-of-sample distributions and paths.

Risk-neutral $\mathbb{Q}$

- Up probability q is implied by u, d, and the risk-free return.
- Used for no-arbitrage pricing and hedging.
- Tested by reproducing traded prices and excluding arbitrage.

A pricing probability is generally not an empirical frequency.

Risk-Neutral Probability Enforces No Arbitrage

Let $R_f > 0$ be the one-step risk-free gross return. For current equity price S_0 and next-step prices $S_0 u$ and $S_0 d$, absence of arbitrage requires:

$$d < R_f < u.$$

The unique risk-neutral up probability that makes the discounted share price a martingale is:

$$S_0 = \frac{q S_0 u + (1-q) S_0 d}{R_f}, \qquad \underbrace{q = \frac{R_f - d}{u - d}}_{\text{risk-neutral probability}} .$$

For a claim paying H_u in the up state and H_d in the down state, the same replication relation gives:

$$V_0 = \frac{q H_u + (1-q) H_d}{R_f}.$$

This is an expectation under $\mathbb{Q}$, not a forecast of the realized payoff.

Probability of Profit Belongs to the Real-World Measure

Let $C_0 > 0$ be a strategy's initial cost, $W_{n,k}$ its terminal wealth at node (n, k), and $R_b > 0$ a benchmark gross return per step. Under $K_n \sim \mathrm{Binomial}(n, p)$, define profit by $W_{n,K_n} > C_0 R_b^n$. Its probability is:

$$\mathrm{POP}_{\mathbb{P}} = \sum_{k=0}^{n} \mathbf{1}_{\{W_{n,k} > C_0 R_b^n\}} \binom{n}{k} p^k (1-p)^{n-k}.$$

POP depends on the forecast law, horizon, benchmark, costs, and profit definition. It is not expected profit and does not report loss severity.

Substituting risk-neutral q would answer a pricing-measure question, not the empirical probability that the strategy succeeds.

Stylized Facts for Binomial Lattice Models

The lattice is our first model of equity-price dynamics. L3a gives three empirical requirements against which to test it.

- Heavy tails: large return magnitudes occur more often than a Gaussian model predicts.
- Weak linear autocorrelation: signed one-step returns usually carry little serial linear dependence.
- Volatility clustering: large movement magnitudes occur near other large movements, and calm observations near calm observations.

The fixed lattice matches the second pattern by assumption, but fails the first and third.

Source: Cont (2001)

Single-Step Growth Is a Bernoulli Transform

Let X_j equal one for an up move and zero for a down move, with $X_j \overset{\text{i.i.d.}}{\sim} \text{Bernoulli}(p)$. The one-step price ratio is:

$$\frac{S_j}{S_{j-1}} = u^{X_j} d^{1-X_j} = d\left(\frac{u}{d}\right)^{X_j}.$$

Let g_j be the continuously compounded growth rate over $\Delta t > 0$ years. Taking logarithms gives:

$$\underbrace{g_j = \frac{1}{\Delta t}\left[\log d + X_j \log\left(\frac{u}{d}\right)\right]}_{\text{two-point growth distribution}}.$$

The lattice growth rate is therefore an affine transformation of a Bernoulli variable and takes only two possible values.

One-Step Mean and Variance

Let g_j be the two-point growth rate, with constant $u > d > 0$, probability $p \in [0, 1]$, and step length $\Delta t > 0$. Its expected value is:

$$\mathbb{E}[g_j] = \underbrace{\frac{p \log u + (1-p) \log d}{\Delta t}}_{\text{mean growth rate}}.$$

Its variance is:

$$\mathrm{Var}(g_j) = \underbrace{\frac{p(1-p)}{(\Delta t)^2} \left[\log\left(\frac{u}{d}\right)\right]^2}_{\text{fixed one-step variance}}.$$

The variance is positive when $0 < p < 1$ and $u \neq d$. It vanishes at a degenerate probability or when the two moves coincide.

Multi-Step Growth Is a Binomial Transform

Let $K_t \sim \mathrm{Binomial}(t, p)$ count the up moves in t steps. The cumulative growth quantity $g_t = (1/\Delta t)\log(S_t/S_0)$ is:

$$\underbrace{g_t = \frac{K_t \log(u/d) + t \log d}{\Delta t}}_{\text{binomial transform}}.$$

Because $\mathbb{E}[K_t] = tp$ and $\mathrm{Var}(K_t) = tp(1-p)$:

$$\mathbb{E}[g_t] = \frac{t}{\Delta t}\left[p \log u + (1-p)\log d\right],$$

$$\mathrm{Var}(g_t) = \frac{tp(1-p)}{(\Delta t)^2}\left[\log\left(\frac{u}{d}\right)\right]^2.$$

The corresponding cumulative log return is $r_t = (\Delta t) g_t = \log(S_t/S_0)$.

Heavy Tails

The fixed binomial lattice does not generate heavy-tailed growth.

- The one-step growth rate takes only two bounded values: $\log u/\Delta t$ and $\log d/\Delta t$.
- The multi-step growth quantity is an affine transformation of the binomial count K_t.
- When $0 < p < 1$ and $u \neq d$, the standardized binomial count converges to a Gaussian distribution as t grows.

The model therefore predicts extreme moves less often than the heavy-tailed empirical distributions examined in L3a.

Vanishing Autocorrelation

Let $g_j = b + aX_j$, where $a = \log(u/d)/\Delta t$, $b = \log d/\Delta t$, and the Bernoulli moves X_j are independent. For every nonzero lag τ:

$$\mathrm{Cov}(g_j, g_{j+\tau}) = a^2\,\mathrm{Cov}(X_j, X_{j+\tau}) = 0.$$

Provided $0 < p < 1$ and $u \neq d$, the one-step variance is positive, so:

$$\underbrace{\rho_g(\tau) = 0 \qquad (\tau \geq 1)}_{\text{one-step autocorrelation}}.$$

The lattice matches weak signed-return autocorrelation by construction. It does not discover that pattern from data: independence is an input assumption.

Nested Windows Create Correlation

Let $g_t = aK_t + bt$ be cumulative growth from step 0 through step t. The windows $0 \to t$ and $0 \to t+\tau$ share their first t moves, so their covariance is not zero:

$$\mathrm{Cov}(g_t, g_{t+\tau}) = a^2 tp(1-p).$$

Using the two variances gives:

$$\underbrace{\rho(\tau) = \sqrt{\frac{t}{t+\tau}}}_{\text{nested-window correlation}} \quad .$$

This correlation is mechanical overlap, not predictability in one-step moves. Nonoverlapping t-step windows have zero correlation under the i.i.d. assumption.

Careful: always state whether a return series uses overlapping or nonoverlapping windows before interpreting its autocorrelation.

Volatility Clustering

The fixed binomial lattice does not generate volatility clustering.

The one-step conditional variance is the same at every node and time:

$$\operatorname{Var}(g_j \mid \mathcal{F}_{j-1}) = \frac{p(1-p)}{(\Delta t)^2}\left[\log\left(\frac{u}{d}\right)\right]^2.$$

Constant u, d, and p create constant conditional volatility. The model cannot produce persistent high- and low-volatility regimes unless those parameters or the move law become state dependent.

Connection to L3a: near-zero signed-return autocorrelation can coexist with strong dependence in squared returns, but the fixed lattice excludes that second dependence.

What the Fixed Lattice Cannot Reproduce

The lattice is valuable because its simplifications are explicit.

- It captures positive prices, a complete finite distribution, weak one-step autocorrelation, and tractable backward valuation.
- It excludes continuous move sizes, heavy tails, volatility clustering, regime changes, and parameter uncertainty.
- A prediction band can have acceptable average coverage and still fail during exactly the extreme or high-volatility periods that matter most.

Use a lattice to expose states, trading rules, and replication. Use richer innovations or state-dependent parameters when the question depends on rare moves or dynamic volatility.

Summary

This lecture built a recombining equity lattice, separated forecasting from pricing measures, and tested the fixed model against empirical return patterns.

Key takeaways

- **Recombination is algebraic.** A node depends on move counts because multiplication commutes. The constraint $ud = 1$ is optional.
- **The measure follows the question.** Real-world p supports forecasts and POP, while risk-neutral q enforces no-arbitrage value.
- **Simple models have visible limits.** The fixed lattice matches weak one-step autocorrelation but misses heavy tails and volatility clustering.

The next step is to turn lattice states into explicit trading and valuation rules.

That’s a wrap. Which lattice assumption would you relax first, and what empirical pattern should the extension recover?